

ABV-Indian Institute of Information Technology and Management Gwalior

Minor Examination (Academic Year 2024-25)

Course Name: Fundamentals of Mathematics and Computer Programming

Date of Examination: 25th September 2025

Timing: 10:00A.M-12:00Noon

(MBA-I Semester)

Maximum Marks: 30

Duration: 2 hours

1. What is the difference between memory read and memory write operations? List the different steps involved in memory read and memory write operations. (6)
2. Differentiate between: (6)
 - a) Main memory and Cache memory
 - b) Magnetic disc and Magnetic tape
 - c) Optical storage and Solid-state devices
3. A university is planning to set up a digital classroom for students. The administration wants to ensure smooth lecture delivery, student engagement, and assessment. Suggest appropriate input and output devices for this classroom and justify your choices. (6)
4. What is the importance of the ALU in computer architecture? (2)
5.
 - a) Convert the decimal number 87.34 to 8421 BCD code.
 - b) Convert the 8421 BCD code 0111010000100110 to the decimal number.
 - c) Convert the decimal number 3456 into XS-3 BCD code.
 - d) Convert the Gray coded number 11010011 to its binary equivalent.
 - e) Convert the binary number 10100011 to its equivalent Gray coded number.
 - f) Convert the binary number 1011.010 into its equivalent in decimal system.
 - g) Convert the hexadecimal number A53 into its equivalent in decimal system.
 - h) Convert the hexadecimal number 6B39 into its equivalent in the decimal system.
 - i) Convert the octal number 5324 into its equivalent in decimal system.
 - j) Convert the decimal number 45796 to its equivalent octal number. (10)

$$\begin{array}{r} 8 \overline{) 665} \\ \underline{64} \\ 25 \\ \underline{24} \\ 1 \end{array}$$

$$\begin{array}{r} 8 \overline{) 5324} \\ \underline{48} \\ 52 \\ \underline{48} \\ 44 \\ \underline{40} \\ 4 \\ \underline{4} \\ 0 \end{array}$$

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$$\begin{array}{r} 8 \overline{) 45796} \\ \underline{40} \\ 57 \\ \underline{48} \\ 97 \\ \underline{80} \\ 17 \\ \underline{16} \\ 1 \\ \underline{0} \\ 1 \\ \underline{0} \\ 1 \\ \underline{0} \\ 1 \end{array}$$

Absoluta

ABV- Indian Institute of Information Technology and Management Gwalior
MBA First Semester
End Term examination- Dec 2025
Subject: FMCP

Time: 3 hrs

MM: 50

Answer any FIVE questions from the followings (5 Nos X 10 Marks)

1. Solve the simultaneous equations through Carmer's Rule

$$2X-3Y+Z=4$$

$$X-3Y+4Z=6$$

$$2X-4Y+4Z=8$$

2. Solve the simultaneous equations through Matrix Method

$$3X+Y-2Z=10$$

$$4x-y+Z=5$$

$$X-3Y-4Z=-1$$

$$-0.8 \times 3 +$$

$$1.6 - 1.2 + 0.4$$

3. Differentiate the following:

a) $y = 5x^3 + 7x^2 - 3x + 9$

b) $y = \frac{1}{x} + 3x^2$

c) $y = \sqrt{x} + \frac{4}{x^2}$

d) $y = (x^2 + 3x)(x - 1)$

e) $y = e^x + \ln x$

MR

AC - TR

4. Solve the following:

a) If $TR = 10Q - 2Q^2$, find MR at $Q = 3$.

b) A firm's total cost function is $TC = 50 + 20Q + 3Q^2$.

Find: (a) MC (b) AC at $Q = 5$. **

c) If $TR = 5Q + 2Q^2$, find AR and MR.

d) Suppose $TC = 100 + 10Q - Q^2 + 0.5Q^3$.

Find MC and AVC. **

e) If $MR = 40 - 4Q$, find TR.

f) Demand: $Q = 100 - 2P$ Find elasticity of demand at $P = 20$.

$$1.25 \times 2 - 3(2.5) + 2.5$$

$$2.5 - 7.5 + 2.5$$

$$2.5 - 5$$

$$\frac{-1.2}{2} = -\frac{1}{2}$$

5. Demand and supply equations:

$$Q_1 + 2Q_2 = 20$$

$$3Q_1 + Q_2 = 30$$

$$\frac{-14}{-1}$$

$$\frac{-1}{2}, \frac{-1}{1}$$

Find the price and quantity demand by using matrix inversion method

$$\frac{-1-2}{2} = -\frac{3}{2}$$

6. A firm faces demand $P = 100 - 2Q$, and total cost $TC = 20Q + 100$.

(a) Derive MR.

(b) Find profit-maximizing output, price, and maximum profit.

$$\frac{-1.2}{-1}$$